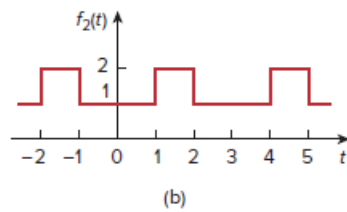


Problem

Plot the amplitude spectrum for the signal $f_2(t)$ in Fig. 17.56(b). Consider the first five terms.

Figure 17.56(b):



Step-by-step solution

Step 1 of 3

This is an even function

$$T = 3, \quad \omega_0 = \frac{2\pi}{3}, \quad b_n = 0$$

$$f(t) = \begin{cases} 1, & 0 < t < 1 \\ 2, & 1 < t < 1.5 \end{cases}$$

$$a_0 = \frac{2}{T} \int_0^T f(t) dt = \frac{2}{3} \left[\int_0^1 1 dt + \int_1^{1.5} 2 dt \right]$$

$$= \left(\frac{2}{3} \right) [1 + 1] = \frac{4}{3}$$

$$a_n = \frac{4}{T} \int_0^T f(t) \cos(n\omega_0 t) dt$$

$$= \frac{4}{3} \left[\int_0^1 1 \cos\left(\frac{2n\pi t}{3}\right) dt + \int_1^{1.5} 2 \cos\left(\frac{2n\pi t}{3}\right) dt \right]$$

$$= \frac{4}{3} \left[\frac{3}{2n\pi} \sin\left(\frac{2n\pi t}{3}\right) \Big|_0^1 + \frac{6}{2n\pi} \sin\left(\frac{2n\pi t}{3}\right) \Big|_1^{1.5} \right]$$

$$= \left[\frac{-2}{n\pi} \right] \sin\left(\frac{2n\pi}{3}\right)$$

Step 2 of 3

$$f_2(t) = \frac{4}{3} - \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{3n\pi}{3}\right) \cos\left(\frac{2n\pi t}{3}\right)$$

$$a_0 = \frac{4}{3} = 1.3333, \quad \omega_0 = \frac{2\pi}{3},$$

$$a_n = \left[\frac{-2}{n\pi} \right] \sin\left(\frac{2n\pi}{3}\right)$$

$$A_n = \sqrt{a_n^2 + b_n^2} = \left| \left[\frac{-2}{n\pi} \right] \sin\left(\frac{2n\pi}{3}\right) \right|$$

$$A_1 = 0.5513, \quad A_2 = 0.2757, \quad A_3 = 0, \quad A_4 = 0.1375, \quad A_5 = 0.1103$$

The Amplitude spectra are shown below

Step 3 of 3

